

Zhai Liang

C++, C#, Python, 2D/3D geometry, Image/video processing expert
AVAILABLE IN ANY TIME ZONE



Email: gbapyou@gmail.com

A freelancer with full experience in software development of over 10 years and especially a deep understanding of 2D/3D geometry and image/video processing. Major programming languages are C++/C#/Python. Most recently, I worked on a video processing project for US company called Prontopia. Also, I worked on some AI projects recently.

SKILL SET

No	Skill	Experience (Years)
1	C++	11
2	C#	7
3	Java	5
4	Python	3
5	PHP	2
6	Visual Studio	11
7	Unity	11
8	Android Studio	7
9	WDK (Windows Driver Kit)	3
10	WinDgb (Windows Kernel Debugger)	2
11	Windows App & Engine Development	7
12	Android App, NDK Development	7
13	Linux App & Engine Development	5
14	Unity Game, App, Plugins Development	5
15	Windows Kernel Module Development	3
16	Reverse Engineering	2
17	Image/video processing	7
18	2D/3D computational geometry	7
19	OpenCV, FFmpeg, Assimp, Glm, OpenFace, Git	7
20	DirectX/OpenGL	2
21	OpenSSL, cURL	2

MAJOR PROJECTS

1. Anti-Cheat System (C++, C#, PHP, cURL, OpenSSL)

<https://github.com/nxtboyIII/fairplay>

A system that protects the mobile game from any kind of hacks. The system has been developed for a game called “BulletForce”, which is an online multiplayer shooting game that is developed by US company called BlayzeGames. I designed a very sophisticated structure in order to confuse the hackers. Also, I modified Photon server client SDK in order to encrypt all data being transferred between players. I used Android NDK (C++ module) for client system, and PHP for the back-end system. Several algorithms like MD5, SHA has been used for secure communication and user validation.

2. Data & Anti-Theft Protection System (C++, ASM, FPGA, VHDL, WinWDK, WinDBG)

A system that protects the data from being stolen. This project is quite similar to TrueCrypt and VeraCrypt, but not exactly the same, because the data is protected in hardware level. We used FPGA as a core firmware device that contains AES-256 algorithm in order to encrypt/decrypt all data being transferred between hard drive and memory. The main board was designed with PCI-X interface. The system works on Windows and Linux. I was responsible of implementing the firmware and Windows version.

3. Automatic Ads Generating Platform (C++, Python, FFmpeg)

A platform that easily generates the Ads videos according to the user input. I have developed this project for a US company, Prontopia (<https://prontopia.com>). I was responsible of designing all structures and implementing the back-end C++ engine that generates the result videos according to the JSON string sent from the user’s input.

4. AI-powered customer-service coordination system (ACS)

The intended users are **logistics or international-trade company** that currently handles customer and supplier coordination through hundreds or thousands of WeChat groups. The core pain point is that customer-service staff manually copy questions between customer groups, supplier groups, and internal teams, then chase answers and send them back. ACS is supposed to automate and track that workflow.

5. Automatic effect/animation for NFT (C++, FFmpeg, Assimp, 2D/3D geometry)

An app that takes images as input and apply effects to generates several animation videos for NFTs. The results are below:

https://drive.google.com/file/d/1wr_5x7ZO2iQ_I9S5MvVljHXXJ16-Vby0/view?usp=sharing

6. Facial expression generating engine (2D/3D geometry)

<https://github.com/basithuss/animate/tree/main>

An Engine that generates the several facial effects like smile, sad, happy, angry from normal face images. This engine is not using any online AI service.

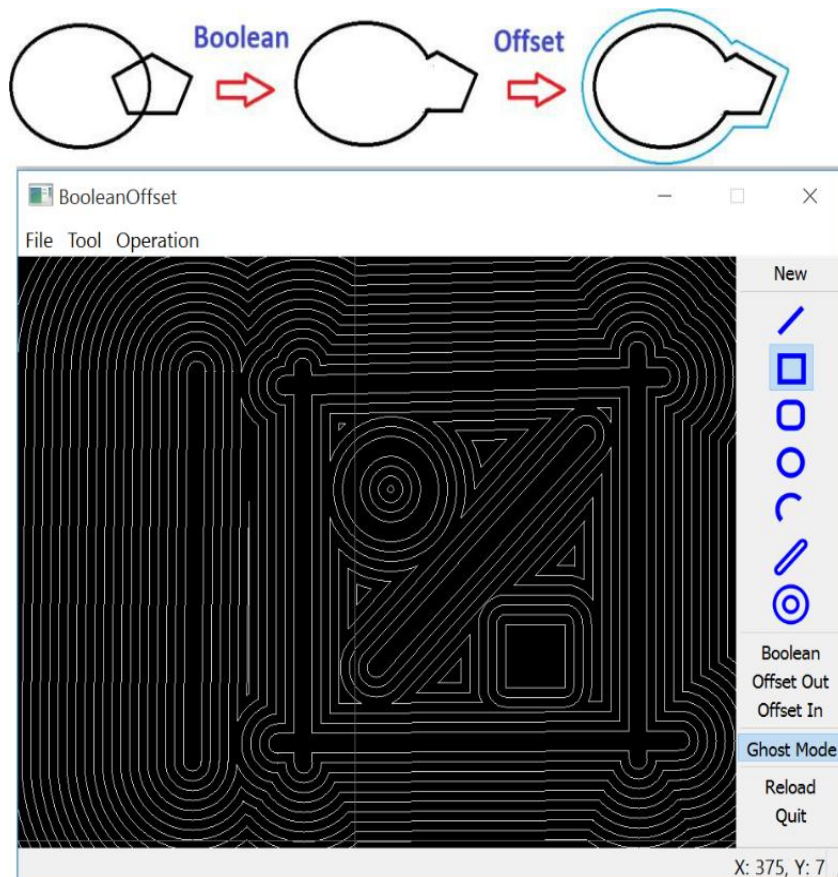


7. Offset/Boolean Operation CAD Engine (C++, QT, Visual Studio, Assimp, 2D geometry).

<https://github.com/Gbapy/Offset-Boolean-CAD>

No third-party libraries and open sources

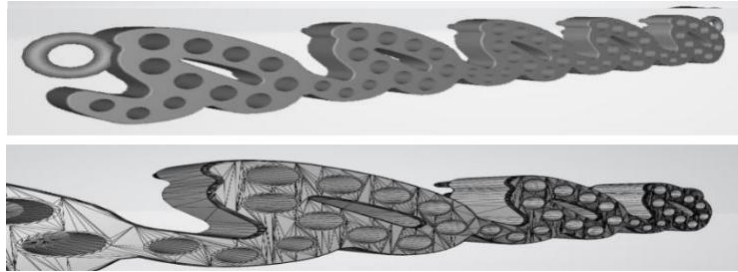
A high-performance CAD engine that performs Offset/Boolean operation, which is very useful for electronic circuit board design. This project has been developed in C++ (Visual Studio & QT) with a lot of advanced data structures and algorithm



8. 3D Model Generating Engine with GLB/STL export for 3D print (C++, Python).

<https://github.com/Gbapy/3D-Model-Generate-Engine-C-Python->
[*No third-party libraries and open sources*](#)

A high-performance back-end engine that generates 3D models of several accessories like rings, necklaces according to user's online input.



This engine is very efficient enough to generate a 3D models in a short time. It only takes about 50 milliseconds to generate a model, even for the very complicated model with vast number of vertices. I used C++ programming language for the core engine (Linux shared object) and Python as a wrapper. This engine supports GLB 2.0/STL export.

9. 3D Modeling Plugin for Unity (C++, C#, Unity, Assimp)

<https://youtu.be/DyM0rXeAc7k>

A plugin for Unity that is used for 3D modeling. An advantage of this plugin is that you can easily generate 3D models from 2D images without having knowledge about tricky things like vertices, meshes, etc.

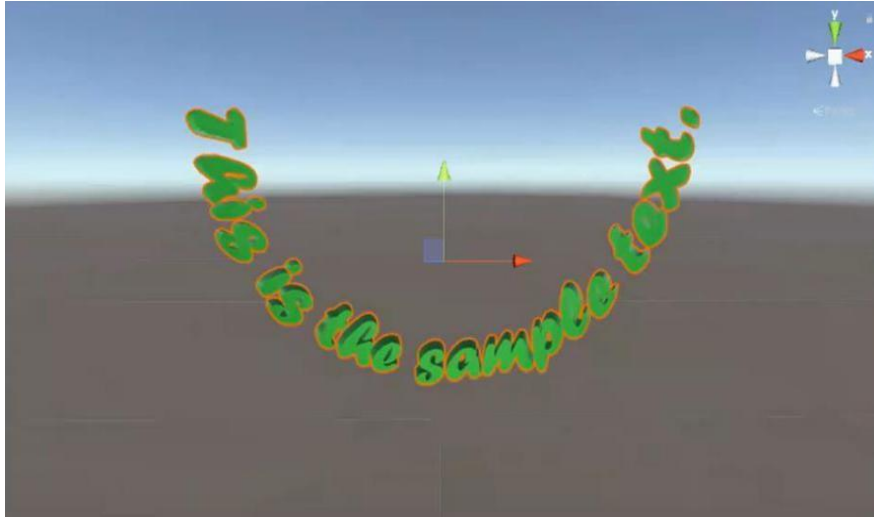


10. Font Plugin for Unity (C++, C#, Unity, Assimp)

<https://github.com/Gbapy/Font-Plugin-for-Unity>

<https://youtu.be/MV9FSutD8Xo>

A plugin for Unity that allows developer to easily generate 3D text models with several fonts ever installed in OS, which makes it easier to handle fonts that is one of very annoying parts of Unity.



11. WeChat bot (C++, Reverse Engineering, IDA pro)

<https://github.com/Gbapy/Wechat-chatbot-using-reverse-engineering>

An engine that hooks WeChat communication module and automatically sends messages (including images) at a certain interval. I used DLL injecting and reverse engineering technologies.

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